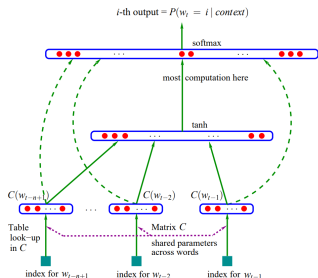


Deep Learning for NLP

Basics

Neural Probabilistic Language Model



Learning goals

- Grasp importance of the “look-up table” a.k.a. embedding layer
- Understand computational implications of language modeling

WHAT IS A LANGUAGE MODEL?

Wikipedia says:

"A statistical language model is a probability distribution over sequences of words"

This means (a) assigning a probability to a sequence of words, e.g.

$$P(\text{"we are all interested in NLP"})$$

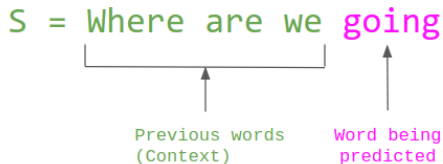
and (b) assigning a probability to the likelihood of a word given a sequence of words, e.g.

$$P(\text{"NLP"} | \text{"we are all interested in"})$$

CF. PREVIOUS CHAPTER I

Predict the next token:

$$\begin{aligned} P_{(w_1, w_2, \dots, w_n)} &= p(w_1)p(w_2|w_1)p(w_3|w_1, w_2)\dots p(w_n|w_1, w_2, \dots, w_{n-1}) \\ &= \prod_{i=1}^n p(w_i|w_1, \dots, w_{i-1}) \end{aligned} \quad (1)$$



$$P(S) = P(\text{Where}) \times P(\text{are} \mid \text{Where}) \times P(\text{we} \mid \text{Where are}) \times P(\text{going} \mid \text{Where are we})$$

► Source: The Gradient

MAKING USE OF THE MARKOV-ASSUMPTION

The Markov-Assumption

- "The future is independent of the past given the present"
- In NLP context:
 - Next word only depends on the k previous words
 - k th order markov assumption with k to be chosen manually

"Traditional" count-based models

- Good baselines, but severe shortcomings
- Lacking the ability to generalize

WHAT ARE POTENTIAL PROBLEMS?

Curse of dimensionality

- Linear increase in context size leads to an exponential increase in the number of parameters
- Considering a vocabulary of size $|V| = 100,000$:
→ Already for bi-grams: $|V|^2 = 10^{10}$ possible combinations

Sparsity

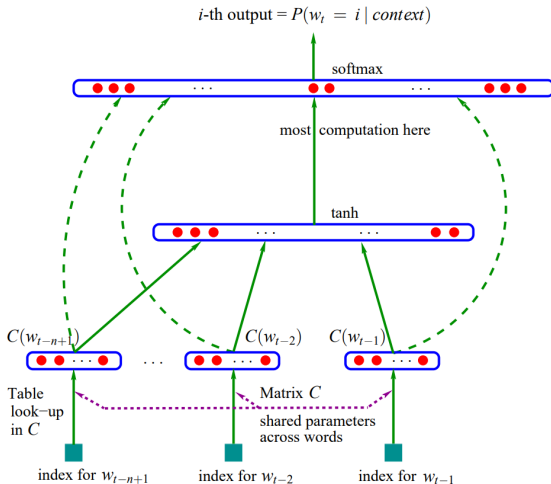
- Again, considering $|V| = 1.000.000$ & bi-grams as context
- Unlikely to observe all of the bi-gram combinations
 - Ⓐ ever
 - Ⓑ often

A NEURAL PROBABILISTIC LANGUAGE MODEL

Idea

- Using a neural network induces non-linearity and overcomes the shortcomings of traditional models
 - Ⓐ Linear increase in #parameters with increasing context size
 - Ⓑ Better generalization
- **Input:** *Context of $(n - 1)$ words* $[w_{(t-n+1):(t-1)}]$
- **In between:**
 - *Look-up table* $[\vec{w}^{(w_{t-n+1})}; \dots; \vec{w}^{(w_{t-2})}; \vec{w}^{(w_{t-1})}]$
 - *Non-linearity* e.g. tanh, ReLU
- **Output:**
Probability distribution over the next word $P(w_t | w_{(t-n+1):(t-1)})$

GRAPHICAL ILLUSTRATION



► Source: Bengio et al., 2003

Note: $C(\cdot)$ replaced by $\vec{w}(\cdot)$ on the previous slide.

WHAT COULD BE PROBLEMATIC?

Computational cost

- Vanilla softmax is expensive
- Proposed solution(s):
 - 1 Hierarchical softmax ► Morin and Bengio, 2005
 - 2 Sampling Approaches (next chapter)

Still relying on the markov assumption

- Context window has to be specified manually